CHAPTER 1

# Solving Quadratics

A quadratic function has three terms: $ax^2+bx+c$. a, b, and c are known as the *coefficients*. The coefficients can be any constant, except that a can never be zero. (If a is zero, it is a linear function, not a quadratic.)

When you have an equation with a quadratic function on one side and a zero on the other, you have a quadratic equation. For example:

$$72x^2 - 12x + 1.2 = 0$$

How can you find the values of x that will make this equation true?

You can always reduce a quadratic equation so that the first coefficient is 1, so that your equation looks like this:

$$x^2 + bx + c = 0$$

For example, if you are asked to solve $4x^2 + 8x - 19 = -2x^2 - 7$

$$4x^2 + 8x - 19 = -2x^2 - 7$$

$$6x^2 + 8x - 12 = 0$$

$$x^2 + \frac{4}{3}x - 2 = 0$$

Here, $b = \frac{4}{3}$ and $c = -2$.

$x^2 + bx + c = 0$ when

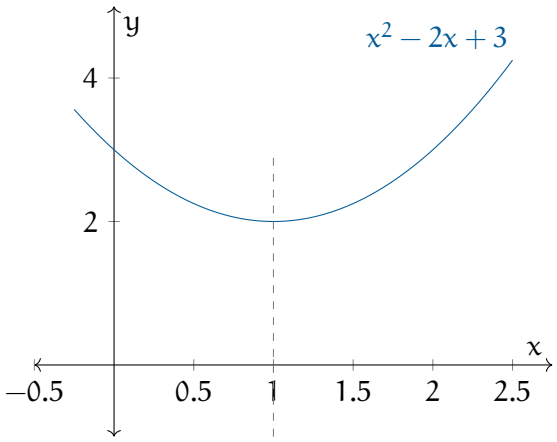
$$x = -\frac{b}{2} \pm \frac{\sqrt{b^2 - 4c}}{2}$$

Only when $a = 1$

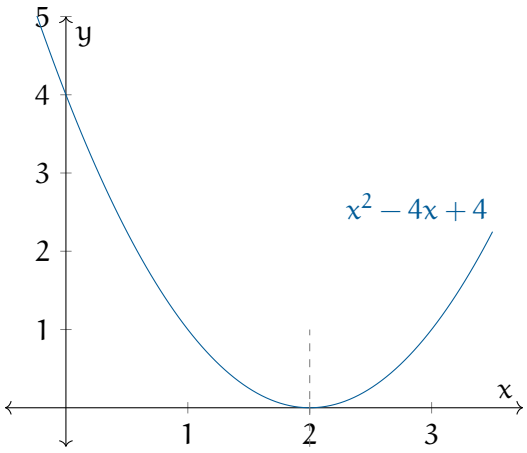
What does this mean?

For any b and c, the graph of $x^2 + bx + c$ is a parabola that goes up on each end. Its low point, the vertex, is at $x = -\frac{b}{2}$. For a quadratic with any value of a, the vertex is at $x = -\frac{b}{2a}$, which is called the vertex formula.

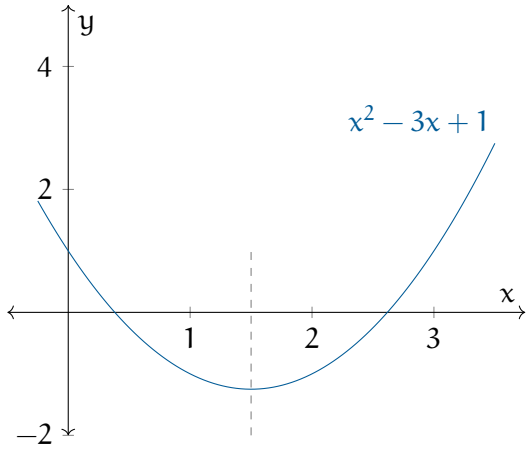
If there are no real roots ($b^2 - 4c < 0$), which means the parabola never gets low enough to cross the x-axis:




If there is one real root ($b^2 - 4c = 0$), it means that the parabola only touches the x-axis.




If there are two real roots ($b^2 - 4c > 0$), it means that the parabola crosses the x-axis twice as it dips below and then returns:




## *Exercise 1* Roots of a Quadratic

*Working Space*

In the last chapter, you found that the function for the height of your flying hammer is:

$$p = -\frac{1}{2}9.8t^2 + 12t + 2$$

At what time will the hammer hit the ground?

*Answer on Page 7*

## 1.1 The Traditional Quadratic Formula

If the last explanation was a little tricky to understand, the quadratic formula is a nifty tool.

**The Quadratic Formula**

$ax^2 + bx + c = 0$ when

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This formula works for any quadratic equation, including ones where $a$ is not 1. Recall the equation from the start of this chapter, $6x^2 + 8x - 12 = 0$. Here $a = 6$, $b = 8$, and $c = -12$. Substitute these into the formula:

$$x = \frac{-8 \pm \sqrt{8^2 - 4(6)(-12)}}{2(6)} = \frac{-8 \pm \sqrt{64 + 288}}{12} = \frac{-8 \pm \sqrt{352}}{12}$$

Since $\sqrt{352} \approx 18.762$, the $+$ gives $x \approx \frac{10.762}{12} \approx 0.897$, and the $-$ gives $x \approx \frac{-26.762}{12} \approx -2.230$.

Watch the signs. Since $c = -12$, the term $-4ac$ becomes $-4(6)(-12) = +288$, so the number under the square root grows.

Earlier, you divided this equation by 6 to get $x^2 + \frac{4}{3}x - 2 = 0$. The formula for $a = 1$ gives the same roots: with $b = \frac{4}{3}$ and $c = -2$, $-\frac{b}{2} \pm \frac{\sqrt{b^2-4c}}{2}$ also comes out to $x \approx 0.897$ and $x \approx -2.230$. Use whichever form is easier for the numbers in front of you.

The formula also answers the question at the start of this chapter. For $72x^2 - 12x + 1.2 = 0$, the number under the square root is $(-12)^2 - 4(72)(1.2) = 144 - 345.6 = -201.6$. No real number is the square root of a negative number, so this equation has no real solutions.

## 1.2 Factoring Quadratics

The quadratic formula always works, but the arithmetic can get heavy, especially without a calculator. For some quadratics, factoring is faster.

Factoring rewrites a quadratic as a product of two simpler expressions. For example, $x^2 + 5x + 6$ can be written as $(x + 2)(x + 3)$. Expanding the product gives back the original: $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$.

The factored form makes the roots easy to read because of the *zero product property*: if two numbers multiply to zero, at least one of them must be zero. So $(x + 2)(x + 3) = 0$ means $x + 2 = 0$ or $x + 3 = 0$, which gives $x = -2$ or $x = -3$. These are the same roots the quadratic formula gives, found without a square root.

Most quadratics you factor by hand have $a = 1$, in the form $x^2 + bx + c$. Certain quadratics with $a \neq 1$ also factor, using the AC method below.

### Types of Factoring Techniques

1. Factoring out the GCF (Greatest Common Factor)
   Always check for a common factor first:

$$2x^2 + 4x = 2x(x + 2)$$

2. Factoring Trinomials: $a = 1$
   For expressions like $x^2 + bx + c$, find two numbers that:
   - Multiply to $c$
   - Add to $b$

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

3. Factoring Trinomials: $a \neq 1$
   Use the "AC method":
   (a) Multiply $a \times c$
   (b) Find two numbers that multiply to $ac$ and add to $b$
   (c) Split the middle term and factor by grouping

$$6x^2 + 11x + 4 = 6x^2 + 3x + 8x + 4 = 3x(2x + 1) + 4(2x + 1) = (3x + 4)(2x + 1)$$

4. Special Cases:
   - Perfect Square Trinomials:
     $a^2 + 2ab + b^2 = (a + b)^2$
     $x^2 + 6x + 9 = (x + 3)^2$
   - Difference of Squares:
     $a^2 - b^2 = (a - b)(a + b)$
     $x^2 - 16 = (x - 4)(x + 4)$

### Tips

- Always factor out the GCF first.
- Check your result by expanding (FOIL). The inverse of factoring is expanding using the FOIL method.
- Some quadratics cannot be factored using integers. For those, use the quadratic formula.

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*This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.*

Appendix A

# Answers to Exercises

## Answer to Exercise 1 (on page 3)

For what $t$ is $-4.9t^2 + 12t + 2 = 0$? Start by dividing both sides of the equation by -4.9.

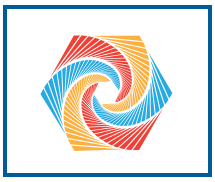
$$t^2 - 2.45t - 0.408 = 0$$

The roots of this are at

$$x = -\frac{b}{2} \pm \frac{\sqrt{b^2 - 4c}}{2} = -\frac{-2.45}{2} \pm \frac{\sqrt{(-2.45)^2 - 4(-0.408)}}{2} = 1.225 \pm 1.382$$

We only care about the root after we release the hammer ($t > 0$).

$1.225 + 1.382 \approx 2.61$ seconds after releasing the hammer, it will hit the ground.



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